



TITLE DEFENSE

**MACHINE LEARNING–BASED SHORT-TERM
FORECASTING OF ELECTRIC VEHICLE
CHARGING STATION LOAD DEMAND**

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BACKGROUND OF THE STUDY

CONTEXT

- Rapid EV adoption increases charging sessions and load variability (Orzechowski et al., 2023)
- Poses challenges to grid reliability and capacity planning
- Accurate short-term forecasting supports efficient grid management
- EV charging load is complex due to session-based, time-varying behavior

STATEMENT OF THE PROBLEM:

The study will address the following questions:

1. How can machine learning-based forecasting models be used to forecast short-term EV charging load demand?
2. How do different machine learning model configurations, SARIMA-GA-TCN and SARIMA-PSO-TCN, perform in forecasting short-term EV charging station load demand?
3. What advantages can be observed from combining different models in terms of improving forecasting reliability?



OBJECTIVES

Specific Objective

1. To preprocess and structure EV charging station data for model training and testing.
2. To model linear and seasonal components using the SARIMA approach.
3. To optimize TCN hyperparameters using the Genetic Algorithm and Particle Swarm Optimization.
4. To predict nonlinear residuals using GA optimized TCN and PSO optimized TCN.
5. To combine the outputs of both models and evaluate performance using MAE, RMSE, SMAPE, and R^2 .
6. To analyze the advantages of combining different modeling approaches in improving forecasting performance and reliability.

SCOPE AND LIMITATION

SCOPE

- Focuses on short-term forecasting
- One week (7 steps ahead) of the future charging demand
- Dataset: City of Boulder EV Charging Station Data (Jan 2018 – Nov 2023).
- Two hybrid models:
 - SARIMA–GA–TCN
 - SARIMA–PSO–TCN

LIMITATION

- Focuses only on data-driven forecasting and model performance.
- Does not cover power quality, grid operation, or energy distribution aspects.



GAP ANALYSIS

Study / Model	Model / Technique	Residual Handling	Optimization	Nonlinear Modeling	Hybrid / Ensemble Structure	Forecasting Scope	Feature Inputs	Evaluation Metrics	Gap / Limitations
SARIMA	Seasonal ARIMA (statistical time series)	✓ (Models residual errors)	✗	✗	✗	Short-term, univariate	Load data only	MAE, RMSE, MAPE	Cannot capture nonlinear relationships; lacks adaptive learning
TCN	Temporal Convolutional Network	✓ (Implicit in convolutions)	✗	✓	✗	Short-term, multivariate	Load, time features	MAE, RMSE	No explicit trend/seasonality modeling; requires hyperparameter tuning
PSO-TCN	TCN optimized via Particle Swarm Optimization	✓	✓ (PSO)	✓	✓	Short-term, multivariate	Weather, load, time	MAE, RMSE, R²	Strong nonlinear capture but lacks decomposition of linear trend components
GA-TCN	TCN optimized via Genetic Algorithm	✓	✓ (GA)	✓	✓	Short-term, multivariate	Weather, time, load	MAE, RMSE, SMAPE	Robust but computationally heavier; lacks explicit seasonal pre-modeling
An Ensemble Methodology for Hierarchical Probabilistic EV Load Forecasting (Buzna et al., 2021)	GBRT + QRF + QRNN Ensemble	✓ (Hierarchical residual correction)	✗	✓	✓	Probabilistic, multi-level	Mainly load data	CRPS, probabilistic score	Not short-term; lacks deep learning temporal modeling
Integrated Approaches in Resilient Hierarchical Load Forecasting via TCN (Türkoğlu et al., 2024)	Temporal Convolutional Network	✓ (Resilient to missing data)	✗	✓	✗	Short-term, residential	Weather, load	MAE, RMSE	No optimization; lacks seasonal-trend extraction
Real-Time Hosting Capacity Assessment for EVs (Zhuang et al., 2025)	Adaptive Spatio-Temporal Graph CNN (ASTGCN)	✓ (Probabilistic error modeling)	✓ (Embedded optimization)	✓	✓	Real-time, probabilistic	Spatial-temporal features	RMSE, MAE, risk	High complexity; not purely forecasting model
Short-Term EV Charging Load Forecasting Based on TCN-LSTM (Tian et al., 2025)	Hybrid TCN + LSTM	✓ (Temporal smoothing)	✗	✓	✓	Short-term, station-level	Weather, similar-day	MAE, RMSE, MAPE	No automatic optimization or explicit seasonality handling
Spatio-Temporal Forecasting of EV Charging Demand (Zhou et al., 2025)	Temporal Graph CNN + Traffic Models	✗	✗	✓	✓	City-level, spatio-temporal	Traffic, user data	MAPE, R²	Focus on spatial, not temporal decomposition; lacks optimization
EV Charging Load Forecasting Using PSO-LSTM (Yang et al., 2025)	LSTM optimized via PSO	✗	✓ (PSO)	✓	✓	Short-term, seasonal	Weather, seasonality	MAE, RMSE, R²	Focused on sequential modeling only; no residual or trend decomposition
SARIMA-GA-TCN	SARIMA + GA-optimized TCN	✓ (SARIMA residuals fed to TCN)	✓ (GA)	✓	✓	7-day-ahead, station-level	Load, weather, time, seasonal	MAE, RMSE, SMAPE, R²	Fills prior gaps: combines decomposition + global optimization for nonlinear learning
SARIMA-PSO-TCN	SARIMA + PSO-optimized TCN	✓ (SARIMA residuals input)	✓ (PSO)	✓	✓	7-day-ahead, station-level	Load, seasonality, user behavior, weather	MAE, RMSE, SMAPE, R²	Enhances model efficiency, reduces training cost, improves temporal precision

GAP ANALYSIS

Key Gaps

GAP 1

Traditional models (SARIMA) capture linear and seasonal patterns but fail to model nonlinear EV charging behaviors

GAP 2

Deep learning models (TCN) can learn nonlinear trends but lack explicit seasonality modeling and need manual hyperparameter tuning.

GAP 3

Optimized models (GA-TCN, PSO-TCN) include tuning methods but miss linear trend decomposition, limiting residual learning accuracy.

GAP 4

Existing hybrid models (e.g., TCN-LSTM) learn complex patterns but lack both automated optimization and structured residual decomposition.

GAP ANALYSIS

Proposed Solution

SOLUTION 1

Combine SARIMA with TCN to handle both linear trends and nonlinear variations for more complete forecasting.

SOLUTION 2

Integrate metaheuristic optimization (GA/PSO) to automate tuning while SARIMA models seasonality.

SOLUTION 3

Use SARIMA decomposition to extract linear trends before applying GA/PSO-optimized TCN for residual learning.

SOLUTION 4

Develop the SARIMA–GA/PSO–TCN framework, combining decomposition and optimization for adaptive short-term EV forecasting.

FRAMEWORK OVERVIEW

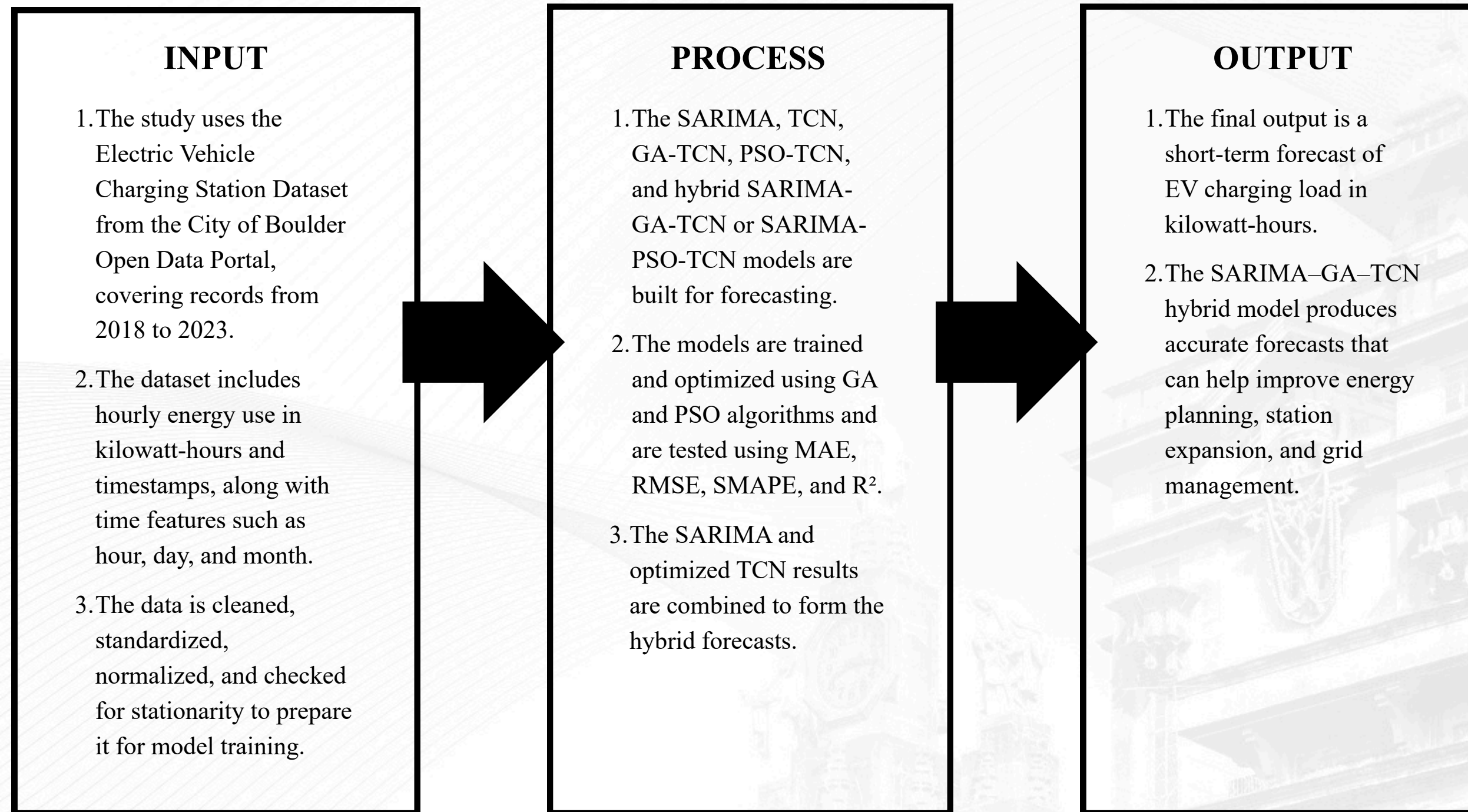


Figure 1.1. : Input-Process-Output Diagram

FRAMEWORK OVERVIEW

INPUT

1. The study uses the Electric Vehicle Charging Station Dataset from the City of Boulder Open Data Portal, covering records from 2018 to 2023.
2. The dataset includes hourly energy use in kilowatt-hours and timestamps, along with time features such as hour, day, and month.
3. The data is cleaned, standardized, normalized, and checked for stationarity to prepare it for model training.

FRAMEWORK OVERVIEW

PROCESS

1. The SARIMA, TCN, GA-TCN, PSO-TCN, and hybrid SARIMA-GA-TCN or SARIMA-PSO-TCN models are built for forecasting.
2. The models are trained and optimized using GA and PSO algorithms and are tested using MAE, RMSE, SMAPE, and R^2 .
3. The SARIMA and optimized TCN results are combined to form the hybrid forecasts.



FRAMEWORK OVERVIEW

OUTPUT

1. The final output is a short-term forecast of EV charging load in kilowatt-hours.
2. The SARIMA–GA–TCN hybrid model produces accurate forecasts that can help improve energy planning, station expansion, and grid management.



METHODOLOGY

Research Design

- Quantitative experimental–comparative research design, involving the comparison of six forecasting models.

Data Preprocessing

- The dataset is the Electric Vehicle Charging Station Dataset from the City of Boulder Open Data Portal. The data covers the period from January 2018 to November 2023.
- Includes hourly energy use in kilowatt-hours and timestamps, along with time features such as hour, day, and month.
- The dataset will undergo data cleaning, timestamp standardization, and normalization
- The dataset is divided chronologically into: 80% for training, 10% for validation, and 10% for testing

METHODOLOGY

Model Development and Implementation

Seasonal Auto Regressive Integrated Moving Average (SARIMA)

- Statistical Forecasting Model
- Captures linear, trend, and seasonal relationships in EV load data.

$$\phi_p(B)\Phi_p(B^S)(1-B)^d(1-B^S)^D X_t = \theta_q(B)\Theta_q(B^S)\omega_t$$

- Output: Predicted load + **residual errors** (nonlinear portion).

Temporal Convolutional Network (TCN)

- Handles nonlinear residuals from SARIMA.
- Uses causal and dilated convolutions to learn temporal dependencies.
- Output: Nonlinear correction to SARIMA forecast.

METHODOLOGY

Model Development and Implementation

Optimization Algorithm

- Mainly Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) will be use to optimize TCN
- Use to automatically tune the hyperparameters :
 - 1.Number of filters
 - 2.Kernel size
 - 3.Learning rate
 - 4.Batch size

Integration

- Merging the output of SARIMA and Optimized TCN

$$\hat{y}_{final(t)} = \hat{y}_{SARIMA(t)} + \hat{r}_{GA-TCN(t)}$$

$$\hat{y}_{final(t)} = \hat{y}_{SARIMA(t)} + \hat{r}_{PSO-TCN(t)}$$



METHODOLOGY

Optimization via GA and PSO

Rationale for Optimization

- Temporal Convolutional Networks (TCNs) require hyperparameter tuning.
 - Metaheuristic algorithms automate this process to minimize forecasting error.

Optimization

- Objective: Automatically adjust TCN hyperparameters (filters, kernel size, learning rate, batch size).
- Goal: Minimize Mean Absolute Error (MAE) — serves as the fitness function.

Genetic Algorithm (GA)	Particle Swarm Optimization (PSO)
Inspired by evolutionary principles.	Inspired by swarm intelligence (e.g., birds/fish).
Uses selection, crossover, and mutation to evolve optimal hyperparameter sets.	Particles update velocity and position using p_{best} (individual best) and g_{best} (global best).
Gradually finds optimal hyperparameters through genetic evolution.	Enhances convergence speed and avoids local minima.

Hybrid Model Process Flow

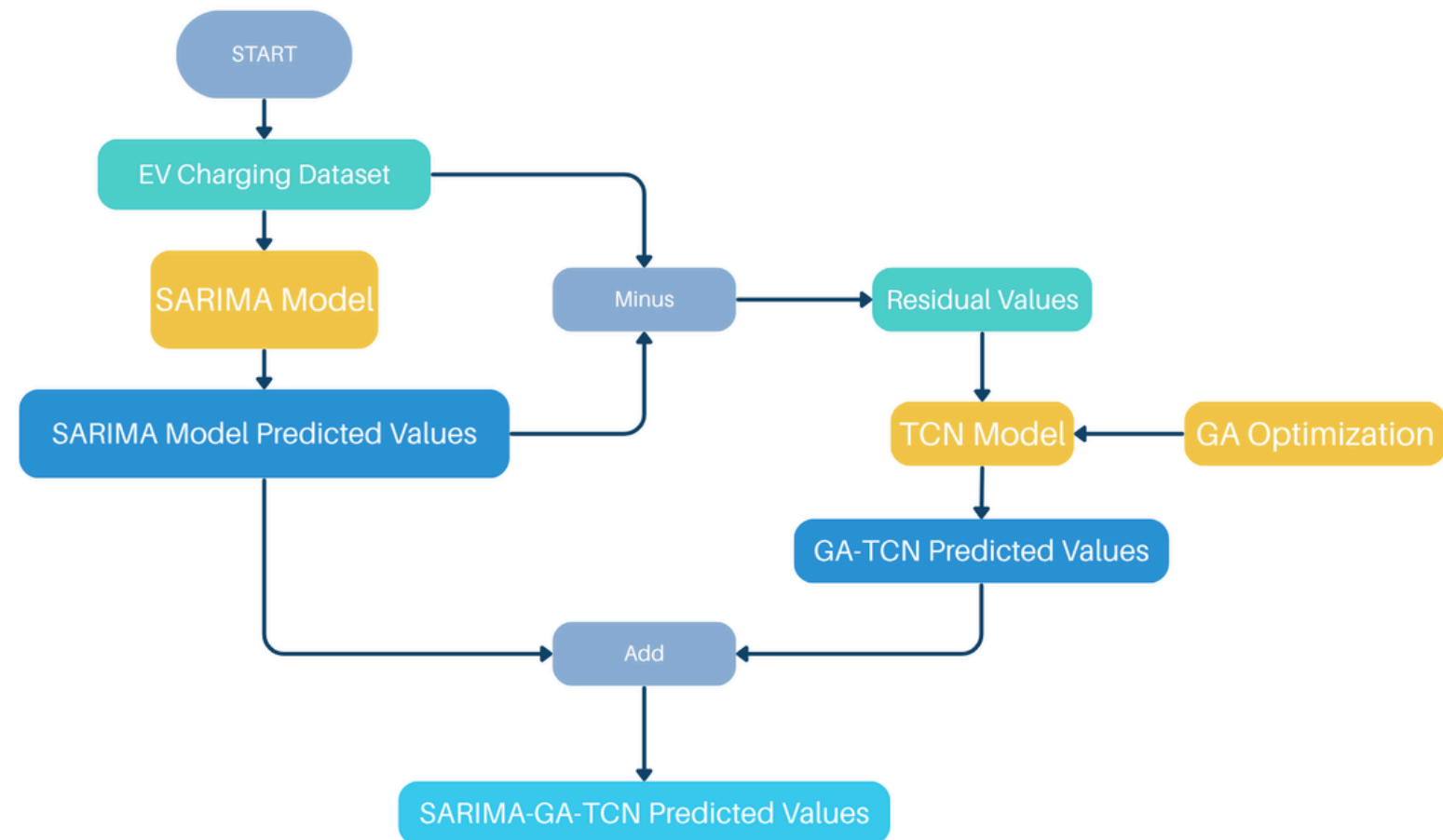


Figure 3.1. : SARIMA-GA-TCN Model

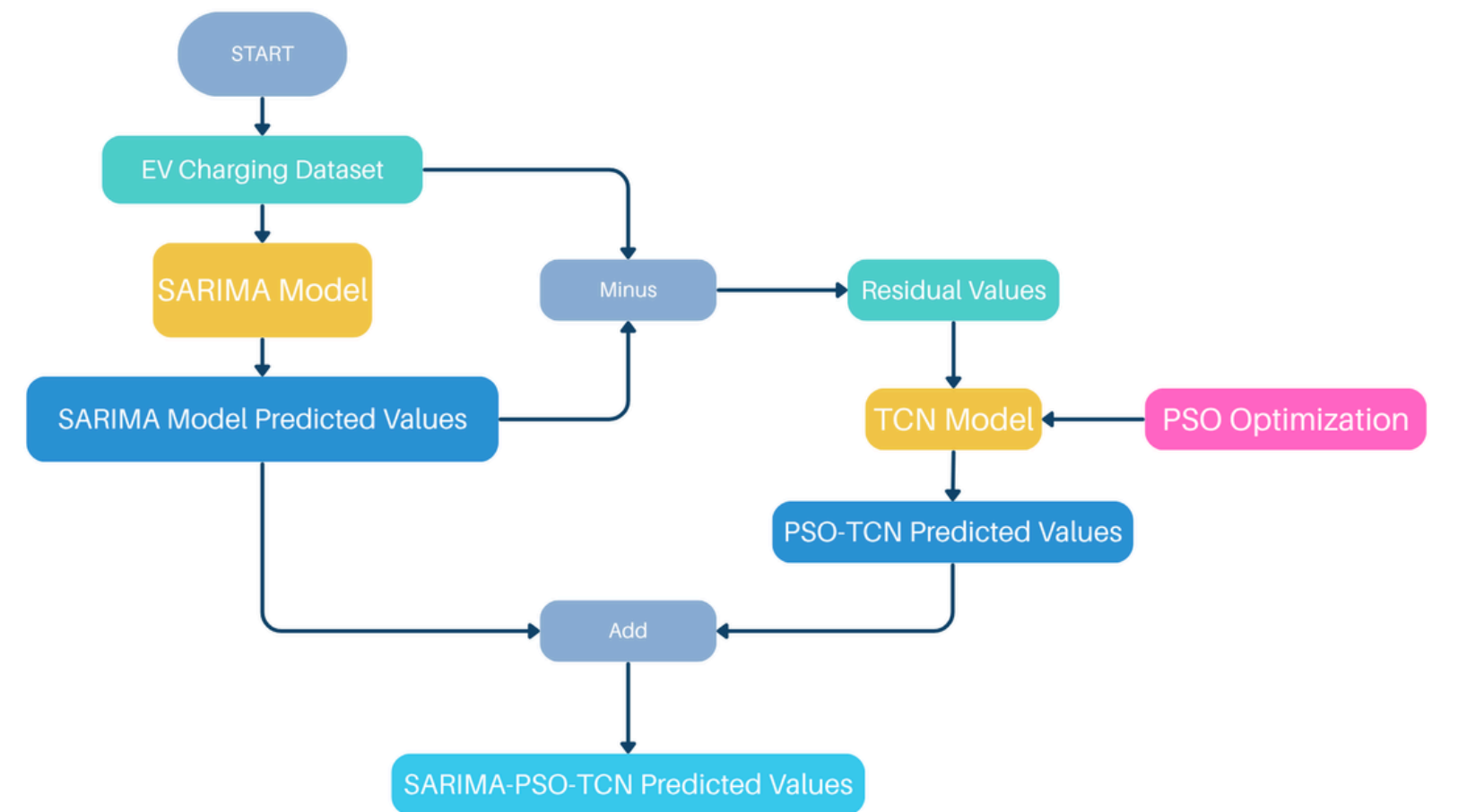


Figure 3.2. : SARIMA-PSO-TCN Model



METHODOLOGY

Evaluation and Comparative Analysis

Evaluation Metrics

- Model performance is assessed using four key statistical indicators:

MAE (Mean Absolute Error)	Measures the average absolute prediction error (kWh).
RMSE (Root Mean Square Error)	Penalizes larger errors, crucial for assessing peak load deviations
SMAPE (Symmetric Mean Absolute Percentage Error)	Provides a balanced percentage error for relative accuracy.
R ² (Coefficient of Determination)	Indicating the model's fit quality — values closer to 1 means better performance.

Comparative Analysis

- The hybrid models are compared against four benchmark configurations: SARIMA, TCN, GA–TCN, and PSO–TCN.
 - Goal: To **analyze the advantages** of combining **decomposition (SARIMA)** and **optimization (GA/PSO)** in enhancing **forecasting reliability**, addressing the gap left by models lacking both components.

SIGNIFICANCE OF THE STUDY

Significance of the Study

Accurate EV load forecasting provides valuable insights for key stakeholders:

1. **Electric Utilities** – Enables better load management, reduces equipment stress, and supports grid expansion and upgrades.
2. **EV Charging Providers** – Ensures efficient operations, optimal resource allocation, and high service reliability.
3. **Researchers** – Offers a hybrid forecasting reference for future studies in energy forecasting and smart grid development.



SUMMARY

The SARIMA-GA-TCN and SARIMA-PSO-TCN frameworks offer a comprehensive and adaptive solution that leverages decomposition and optimized deep learning to achieve highly accurate short-term EV charging demand forecasts, addressing critical gaps in current methodologies.



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THANK YOU!